

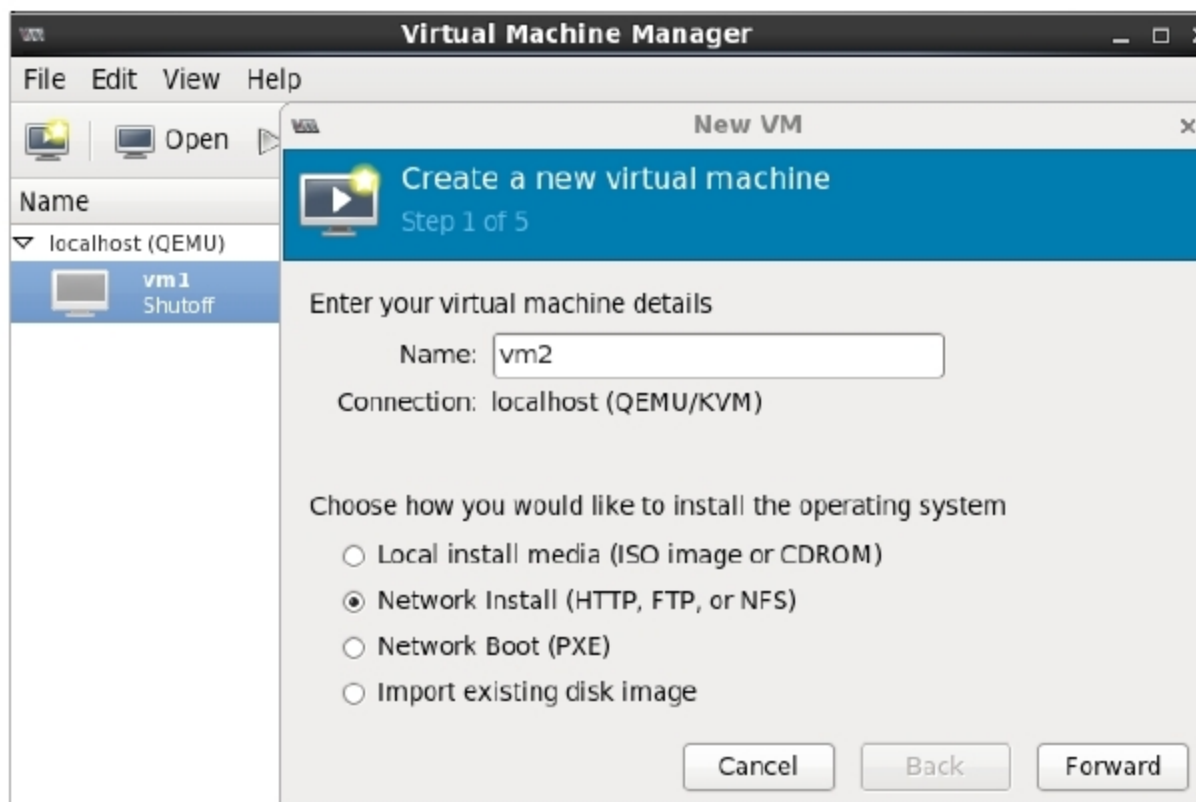


Figure 2: Naming the VM

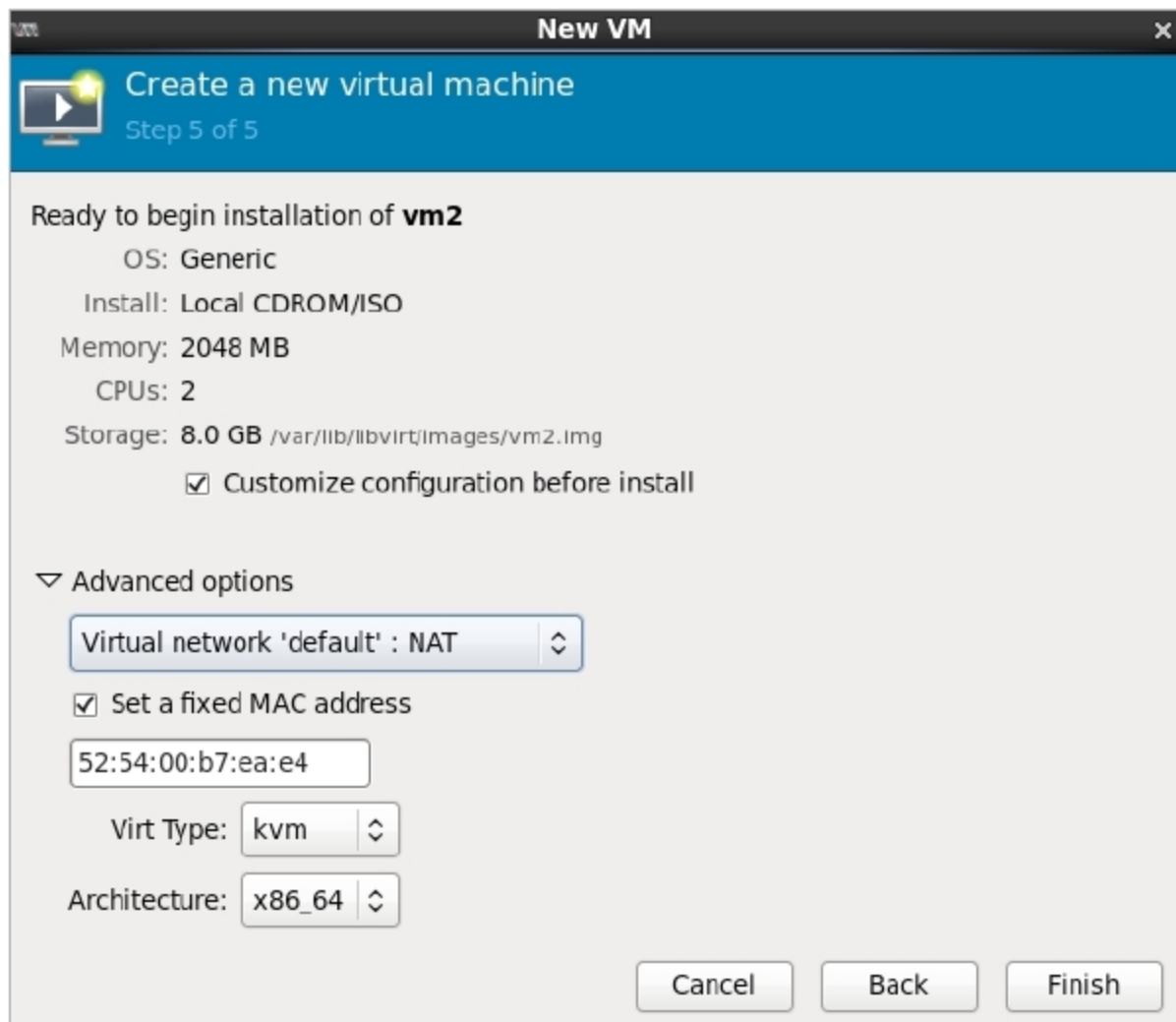


Figure 3: VM networking configuration

Step 2: Install guest VM by clicking on 'new', then enter the name of your guest machine. In my test-bed, it is named 'vm2' to create the instance of an Ubuntu VM. I am also using the NFS server to install the guest OS shown in Figure 2.

Step 3: To eliminate the overhead of the VM's disk storage during the migration process, implement the NFS server which stores the guest OS virtual machine's .iso image.

Step 4: The VM is configured with a range of RAM from 1GB to 8GB and

Network Address Translation (NAT) as a network interface. A screenshot of the VM with RAM of 2GB and NAT configuration is given in Figure 3.

The purpose of NAT networking configuration for virtual machines is that it enables network access from the host operating system machine.

Step 5: Connect to the destination remote server.

The connectivity from the source server to the destination with IP address 172.20.0.3 is made using the Secure Shell (SSH) method. Now, click on 'begin installation'; the remaining steps are the same as those of the standard installation of Ubuntu OS, as shown in Figure 4.

Migrating the virtual machine using QEMU-KVM hypervisor

The steps to perform VM migration using KVM hypervisor are as follows.

Step 1: A secure SSH link is set up between the source and the target server.

Step 2: A VM is started at the source host, and the channel bandwidth parameter is configured for the QEMU-KVM hypervisor. The list of the newly deployed virtual machines is shown in Figure 5, and it can be checked using the following command:

```
$ virsh list - all
```

Step 3: Cloud workload benchmarks are executed on the VMs. Now initiate the VM migration process using the following command, as shown in Figure 6:

```
$ virsh migrate -- live vm2 qemu+ssh://server3.eexample.com/system
```

Future of migration technology in cloud computing

VM migration is a fundamental operation for cloud computing virtualisation. In a cloud computing